Experiment No. 2

Date: / /

Aim: To determine the concentration in terms of molarity of $KMnO_4$ solution by titrating it against standard solution of ferrous ammonium sulphate.(Mohr's salt).

Theory: The substances available in the state of high purity are used to prepare standard solution by dissolving a fixed/definite mass in definite volume of distilled water. The molar mass of ferrous ammonium sulphate is 392 g/mol.

A. Preparation of standard solution of ferrous ammonium sulphate (F.A.S.)

Molar mass of ferrous ammonium sulphate (Mohr's salt) ($FeSO_4.(NH_4)_2SO_4.6H_2O$) is 392 g/mol

For 1000 mL 1 M Mohr's salt required mass is 392 g of Mohr's salt.

Hence for 100 mL of 0.1 M Mohr's salt the required mass is $= \frac{392 \times 100 \times 0.1}{1000}$

$= 3.92$ g of Mohr's salt

Apparatus: standard flask 100 mL, balance, watch glass, beaker, glass rod, etc.

Chemicals: Ferrous ammonium sulphate, distilled water, sulphuric acid.

Procedure :

1. Weigh accurately 3.92 g Ferrous ammonium sulphate on watch glass.
2. Transfer the weighed F.A.S. to a beaker and wash the watch glass with distilled water and transfer washings to the beaker. Add little distilled water to dissolve it by stirring.
3. Transfer the solution of F.A.S. from beaker to 100 mL standard flask. Wash the beaker twice with distilled water and transfer washings to the 100 mL standard flask. Add few drops of H_2SO_4 to get clear solution. Dilute the solution up to the mark on standard flask to make volume 100 mL.

B. Determination of molarity of $KMnO_4$ solution by using standard solution of F.A.S.

Apparatus: Burette, pipette, conical flask, white porcelain tile, etc.

Chemicals: $KMnO_4$ solution, 0.1 M F.A.S., 2M H_2SO_4.

Procedure:

1. Wash the burette, pipette and conical flask with water.
2. Rinse the burette with given $KMnO_4$ solution and then fill it. Remove air bubble from the nozzle and avoid leakage if any.
3. Adjust the level of $KMnO_4$ solution in burette up to zero mark with upper meniscus and fix it on a burette stand.
4. Rinse the pipette with standard 0.1 M F.A.S. solution.
5. Pipette out 10 mL of 0.1M F.A.S. solution and transfer it into a clean conical flask. Then add one test tube 2M sulphuric acid. (The solution remains colourless)
6. Place the conical flask on a white porcelain tile below the burette.
7. Start adding $KMnO_4$ solution drop wise from burette in a conical flask with continuous stirring till light/faint pink colour is obtained in conical flask. Keep the flask constant shaking in a circular manner. This is called swirling of solution. See that the colour does not disappears even on vigorous shaking. This is end point of titration.
8. Note this reading as 'Pilot reading' (pilot reading is always in whole number.)
9. Now fill the burette again with $KMnO_4$ solution up to the zero mark with upper meniscus.
10. Repeat the above procedure and take minimum three more correct burette readings.
11. Note down the constant burette reading ....(x) mL (C.B.R.)
12. From C.B.R. calculate the molarity of $KMnO_4$ solution.

Observations:

1. Solution in a burette : $KMnO_4$
2. Solution by a pipette : FAS solution (10 mL) + full test tube of dil. H_2SO_4
3. Solution in conical flask : FAS + dil. H_2SO_4
4. Indicator : $KMnO_4$ itself
5. End point : Colourless to faint pink
6. Chemical Equation : $2KMnO_4 + 10FeSO_4(NH_4)_2SO_4 + 8H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 5Fe_2(SO_4)_3 + 10(NH_4)_2SO_4 + 8H_2O$

Observation Table:

Burette level	Pilot reading	Burette reading in mL			C.B.R.
		I	II	III	
Final	10.0	10.5	10.5	10.5	10.5 (x) mL
Initial	to 0.0	0.0	0.0	0.0	
Difference	mL	10.5	10.5	10.5	

Calculation: From above chemical equation

$$2KMnO_4 + 10FeSO_4(NH_3)_2SO_4 + 8H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 5Fe_2(SO_4)_3 + 10(NH_3)_2SO_4 +$$

2 moles 10 moles

$2 \times 158 = 10$ moles of mohr's salt (molar mass of $KMnO_4 = 158$ g)

$\therefore$ 1000 mL 10M Mohr's salt $\equiv$ 316 g of $KMnO_4$

$$\therefore 10 \text{ mL } 0.1\text{M Mohr's salt} = \frac{316 \times 10 \times 0.1}{1000 \times 10}$$

$$= 0.0316 \text{ g of } KMnO_4$$

$\therefore$ Hence .....(x) CBR mL of $KMnO_4$ solution contains = 0.0316 g of $KMnO_4$

$$\therefore 1000 \text{ mL of } KMnO_4 \text{ contains} = \frac{0.0316 \times 1000}{10.5 \text{ (x) CBR} \times 158}$$

Hence molarity of $KMnO_4$ in the solution is $= \dfrac{0.0316 \times 1000}{10.5 \text{ (x) CBR} \times 158}$

$$= \frac{0.2}{10.5 \text{ (x) CBR}}$$

$$= \frac{0.2}{10.5} \text{ M}$$

$$= 0.0190 \text{ M} \approx 0.02 \text{ M}$$

Space for log calculation

Number	log
10.5	1.0212

Result: The molarity of $KMnO_4$ solution is ....0.02..........M

Remark and sign of teacher: ....
....
....

MCQ

Select [✓] the most appropriate answer from given alternatives of each sub question.

1. The indicator used in Redox titration is----
   - a. phenolphthalein
   - b. methyl orange
   - c. methylene blue
   - ✓ d. potassium permanganate

2. Select the titrant used in redox titration
   - a. oxalic acid
   - b. F.A.S
   - ✓ c. $KMnO_4$
   - d. H_2SO_4

3. The oxidation state of Mn in $KMnO_4$ in redox titration before titration is +7, the oxidation state of Mn after completion of titration is----
   - ✓ a. +2
   - b. +3
   - c. +4
   - d. +7

4. The oxidation state of Iron in Mohr's salt is +2, the oxidation state of Iron after completion of titration is-----
   - a. +1
   - b. +2
   - ✓ c. +3
   - d. +4

5. The role of Mn^{2+} ion during oxidation-reduction titration of $KMnO_4$ by oxalic acid is-----
   - a. catalyst
   - b. reductant
   - ✓ c. oxidant
   - d. reactant

Short answer questions

1. What specific name is given to the permanganate titrations?

Ans. Redox titrations involving potassium permanganate are called permanganometric titrations. In these reactions, MnO_4^- ion acts as the self indicator.

2. Why heating is not required in F.A.S. and potassium permanganate titration?

Ans. Because reaction rate is very high even at room temperature.

3. Why is dilute sulphuric acid added while preparing a standard solution of Mohr's salt?

Ans. To prevent the hydrolysis of ferrous sulphate.

4. Calculate the molar mass of Mohr's salt (Fe = 56, S = 32, N = 14, H = 1, O = 16)

Ans. Molar mass of $FeSO_4(NH_4)_2SO_4 \cdot 6H_2O$

$= (1\times56) + (2\times32) + (2\times14) + (14\times1) + (20\times16)$

$= 56 + 64 + 28 + 14 + 320$

$= 482\ g\ mol^{-1}$

Hence, molar mass $= 482\ g\ mol^{-1}$.

5. Calculate the amount of F.A.S. required to prepare M/20 standard solution of F.A.S.

Ans. Molar mass of FAS $(NH_4)_2Fe(SO_4)_2 \cdot 6H_2O = 392\ g\ mol^{-1}$

Molarity required $= \frac{1}{20}$ M, Volume = 1 L

Mass of FAS = M × Molar mass × Volume

$= \frac{1}{20} \times 392 \times 1$

$= 19.6\ g$

Hence, 19.6 g of FAS is required to prepare M/20 (0.05 M) standard solution.

Remark and sign of teacher: ....

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